

# A $\delta^2$ – $L^3$ Sensitivity Law for Finite-Window Weil Positivity, with a Function-Field Control of the Connes–van Suijlekom Pipeline

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## Abstract

The Connes–van Suijlekom and Connes–Consani–Moscovici programs reduce the Riemann hypothesis to spectral properties of a quadratic form restricted to test functions supported in a finite window. We ask how sensitive that form is to a zero moved off the critical line, and we calibrate the answer in a setting where the truth is known: curves over finite fields, where the window form is exactly the Frobenius Gram matrix of Hallouin–Perret and the Riemann hypothesis is Weil’s theorem. Our main result is a sensitivity law. In the function field we prove that moving a conjugate pair off the critical circle to radius  $\rho$  gives  $\lambda_{\min}(T_R) = -\binom{R+2}{3}a^2 + O(a^4)$  with  $a = \log \rho$ , the coefficient being twice the centred second moment of the window. On the zeta side we prove an unconditional bound  $\lambda_{\min}(Q_\delta) \geq -c(L)\delta^2 - O(\delta^4)$  with  $c(L) = 8L^3(\frac{1}{3} + \frac{1}{\sqrt{5}})$ , valid on windows where positivity is certified, and a sharper conditional bound with constant  $8L^3/3$ ; numerically the re-optimised minimiser attains 94.5% of the sharp large-ordinate constant  $4L^3/3$ . Under the identification  $R \leftrightarrow 2L$  the two laws are the same expression, (window length) $^3/6$ . We also run the full Connes–van Suijlekom minimiser pipeline on deliberately planted off-circle spectra, which separates the steps that carry arithmetic content from those that do not. **Nothing here proves or advances the Riemann hypothesis for  $\zeta$ .**

## Non-claims

Stated first, not in an appendix.

1. **No proof or advance of RH for  $\zeta$ .** Part I runs where the Riemann hypothesis is a theorem (Weil, curves over finite fields); Part II uses fabricated spectra.
2. **No new spectral architecture.** The window form, its Toeplitz structure and its Carathéodory–Fejér theory are due to Hallouin–Perret [8] and Connes–van Suijlekom [5]; the finite-window Weil form itself goes back to Yoshida [10] and Bombieri [1].
3. **Planted spectra are fabricated diagnostics**, not  $L$ -functions of anything.
4. **Numerical detections are tier T2 unless certified**, and *non-detections certify nothing*: a finite basis that finds no negative direction is not evidence of positivity. This asymmetry is the subject of §6.
5.  $\lambda^*(0.8) \leq 2.2702 \times 10^{-17}$  **is a variational upper bound** from a finite basis, not a certified enclosure; we have no rigorous lower bound.
6. **§5 reproduces published computations** (Groskin [6]) at smaller scale.

7. **Block F is claimed only for the function field.** On  $\zeta$  windows the indefiniteness of  $A$  and  $M$  depends on where  $-\log \pi$  is placed; see §7 and Retraction 15.
8. **Not attempted:** Davenport–Heilbronn; any Lean formalisation.

## Epistemic tiers

| Tier | Meaning         | Discharged by                                                  |
|------|-----------------|----------------------------------------------------------------|
| T1   | exact or proved | a proof, or exact arithmetic over $\mathbb{Z}$ or $\mathbb{Q}$ |
| T2   | numerical       | stated working precision + a convergence or noise-floor check  |
| T3   | interpretation  | argument from T1/T2 evidence, labelled as such                 |

Every numbered statement below carries a tier, the script that produced it and the data file that records it. All numerics use `mpmath` at the stated precision; `float64` is used only as a deliberate contrast (§10).

## 1 The pipeline and the question

Let  $X/\mathbb{F}_q$  be a smooth projective absolutely irreducible curve of genus  $g$ , and let  $N_n = \#X(\mathbb{F}_{q^n})$ . Hallouin and Perret [8, Prop. 5] show that the normalised classes  $\gamma^k$  of the iterated Frobenius graphs have Gram matrix

$$\text{Gram}(\gamma^0, \dots, \gamma^R) = (t(|i - j|))_{0 \leq i, j \leq R}, \quad t(0) = 2g, \quad t(n) = \frac{(q^n + 1) - N_n}{q^{n/2}}, \quad (1)$$

a symmetric Toeplitz matrix, which we write  $T_R$ . We adopt (1) as the definition of the window form; it is *their* matrix, entry for entry.

The Connes–van Suijlekom pipeline [5] passes through four steps: (i) reality of the zeros of the ground-state polynomial; (ii) the sign of  $\lambda_{\min}$ ; (iii) simplicity and evenness of the ground state; (iv) convergence of the normalised minimiser. The question of this note is *quantitative*: how large a negative eigenvalue does a zero moved a distance  $\delta$  off the critical line actually produce, and how large a window is needed to see it?

**What is claimed as new.** Only two things.

(N1) The  $\delta^2$ – $L^3$  sensitivity law: Theorem 6 (function field), Theorems 7 and 8 ( $\zeta$ ), the measured near-saturation, and the shared (window length)<sup>3</sup>/6 form.

(N2) The planted-spectrum control of the CvS minimiser pipeline in the function field (§3).

Everything else is foundation or confirmation; §8 says exactly whose.

## 2 Foundation (prior art, stated as such)

**Proposition 1** (Hallouin–Perret [8, Prop. 5, §1.2]). *[T1, cited] The Gram matrix of the normalised iterated-Frobenius classes is the Toeplitz matrix (1).*

**Theorem 2** (Hallouin–Perret [8, Thm. 6, §1.3]). *[T1, cited] Let  $d$  be the rank of the Frobenius space  $F$ . Then  $d$  is the minimal integer such that  $\text{Gram}(\gamma^0, \dots, \gamma^d)$  is singular, a generator of that kernel exists over  $\mathbb{Q}$ , and the resulting Frobenius eigenvalues are algebraic integers of modulus  $\sqrt{q}$ .*

**Theorem 3** (Hallouin–Perret [8, Thm. 36(ii), App. A.2]). *[T1, cited] If a real symmetric positive semi-definite Toeplitz matrix of size  $n + 1$  has rank  $n$  and kernel vector  $(a_0, \dots, a_n)$ , then  $P = \sum a_j X^j$  has  $n$  distinct roots, all of modulus 1.*

Theorem 3 is step (i) of the pipeline; it is the Carathéodory–Fejér corollary that CvS [5, Cor. 1.1] prove again  $C^*$ -algebraically. Their Lemma 33 (rank = largest non-zero *leading* minor) and Lemma 34 (the  $(\gamma_i \pm \gamma_{n-i})/\sqrt{2}$  parity split) are the computational tools we use throughout. CvS themselves point to [8] for the function-field connection and write that “the key difficulty in this context, then, becomes the verification that zero is indeed the (simple) minimal eigenvalue of  $T$ ” [5, Introduction].

### 3 The control experiment

#### 3.1 Genuine spectra

[T1 for the arithmetic, T2 for the spectra; `src/stage1.py`, `results/stage1_spectra.csv`] Four curves ( $y^2 = x^3 + x/\mathbb{F}_5$ ,  $y^2 = x^5 + x^2 + 1/\mathbb{F}_3$ ,  $y^2 = x^7 + x + 1/\mathbb{F}_5$ ,  $y^2 = x^7 + x^3 + x + 1/\mathbb{F}_3$ ;  $g = 1, 2, 3$ ) with brute-force point counts,  $L$ -polynomials by Newton’s identities plus the functional equation, and exact integer power sums. All 40 rows have  $\lambda_{\min} \geq 0$ ;  $\lambda_{\min}(R)$  decreases monotonically (Cauchy interlacing) and vanishes exactly at  $R = d$ , where the kernel polynomial’s roots are the  $e^{i\theta_j}$  — Theorem 2 in action.

#### 3.2 Planted spectra

[T1, exact over  $\mathbb{Q}$ ; `src/stage2.py`, `src/exact_inertia.py`] We replace the Frobenius spectrum by a functional-equation-closed multiset with  $|\beta| \neq 1$ . Choosing rational  $\rho$  and  $\cos \varphi$  makes every  $t(n)$  rational, so the inertia is certified *exactly* by the Sylvester–Jacobi leading-minor rule.

| configuration   | planted                     | first $R$ with $\lambda_{\min} < 0$ | step (i) holds? | $\lambda_{\min}$ simple? |
|-----------------|-----------------------------|-------------------------------------|-----------------|--------------------------|
| on-line control | $ \beta  = 1$               | never (PSD, proved)                 | always          | fails for $R \geq 5$     |
| real pair       | $\rho = 1.3$                | $R = 1$                             | always          | never fails              |
| quartet         | $\rho = 1.3, \varphi = 0.7$ | $R = 2$                             | always          | never fails              |
| two quartets    | $\rho = 1.3$                | $R = 4$                             | always          | never fails              |
| mixed           | quartet + pair              | $R = 4$                             | always          | never fails              |

**Observation 4.** [T1 reasoning, T2 evidence] Step (i) holds at every applicable row of every planted configuration, including rows where  $\lambda_{\min} = -642.8$ : measured  $\max||z| - 1| \leq 5.1 \times 10^{-48}$ . Steps (i) and (iii) are therefore free of arithmetic content. This is not a discovery — CvS state Theorem 3 for an arbitrary real even distribution, so no arithmetic can enter step (i) — but the planted control makes it concrete, with numbers attached.

#### 3.3 Degeneracy beyond the window degree

[T1 corollary of Thm. 3(i); `src/function_level.py`] Since the shift has  $P$  as both minimal and characteristic polynomial,  $\ker T_R$  is the ideal generated by  $P$  truncated at degree  $R$ :

$$\ker T_R = \{P \cdot q : \deg q \leq R - d\}, \quad \dim \ker T_R = \max(0, R + 1 - d), \quad (2)$$

matched in every computed row. Hence  $\lambda_{\min} = 0$  is simple exactly at  $R = d$  and never after, so the CvS simple-ground-state hypothesis fails for every  $R > d$ ; and since  $P$  is self-reciprocal, the kernel splits into even and odd parts of dimensions  $\lceil \frac{m+1}{2} \rceil, \lfloor \frac{m+1}{2} \rfloor$  with  $m = R - d$ , both non-zero for  $m \geq 1$ . CvS state this in words in their Remark 2.3; we record the dimensions.

### 3.4 Step (iv) at the function level

[T1; src/function\_level.py]

**Proposition 5.** *If  $\lambda_{\min}$  is simple for infinitely many  $R$  and the normalised ground-state polynomials converge locally uniformly to  $F \not\equiv 0$ , then every zero of  $F$  lies on  $|z| = 1$  by Theorem 3 and Hurwitz’s theorem. Hence they cannot converge to a target with a zero off the circle.*

The free witness is exactly the obstruction. Measured: zero *angles* converge to  $\sim 10^{-3}$  even for off-line spectra, while the coefficient distance to the target is exactly  $\sqrt{2}$  (orthogonal) and the normalised  $|\hat{c}_R(\beta)|$  *rises* to 0.58–0.71 out to  $R = 36$  instead of falling. This makes concrete the Hurwitz step that is CvS’s own Step 5.

## 4 The $\delta^2$ – $L^3$ law

### 4.1 Function field

**Theorem 6.** [T1; proof in notes/proofs.md, checked by src/proofs\_check.py] *Let the normalised spectrum be a single off-circle real pair  $\{\rho, \rho^{-1}\}$  and put  $a = \log \rho$ . With  $u_i = \rho^i$ ,  $w_i = \rho^{-i}$  ( $0 \leq i \leq R$ ),*

$$T_R = uw^\top + wu^\top, \quad \lambda_{\min}(T_R) = (R+1) - \|u\| \|w\|,$$

*all other eigenvalues being 0. Consequently  $\lambda_{\min} < 0$  for every  $R \geq 1$  as soon as  $\rho \neq 1$ , by strict Cauchy–Schwarz;  $\lambda_{\min}$  is an even function of  $a$ ; and*

$$\lambda_{\min}(T_R) = -2a^2 \sum_{i=0}^R \left(i - \frac{R}{2}\right)^2 + O(a^4) = -\binom{R+2}{3} a^2 + O(a^4).$$

*For  $R = 1$  this is exact:  $\lambda_{\min} = -4 \sinh^2(a/2)$ .*

The coefficient is twice the *centred* second moment of the window; the centring falls out of the optimisation, not from a convention. Checks: the factorisation to  $6.2 \times 10^{-61}$ , the eigenvalues to  $1.0 \times 10^{-59}$ , the moment identity exactly for  $R = 1, \dots, 199$ , and  $-\lambda_{\min}/a^2 = \binom{R+2}{3}$  to twelve printed digits for  $R = 1, 2, 4, 8, 16, 32$ .

### 4.2 Zeta side

Let  $f$  be real, even, supported in  $[-L, L]$ ,  $\|f\|_2 = 1$ , and  $F = \hat{f}$ . Moving a double on-line zero at ordinate  $\gamma$  to  $\frac{1}{2} + \delta \pm i\gamma$  — the count-preserving counterfactual, and the only one continuous as  $\delta \rightarrow 0$  — changes the form by

$$Q_\delta(f) - Q_0(f) = -4\delta^2(F'(\gamma)^2 + F(\gamma)F''(\gamma)) + R_4. \quad (3)$$

**Theorem 7** (unconditional on certified windows). [T1] *With  $c(L) = 8L^3(\frac{1}{3} + \frac{1}{\sqrt{5}}) \approx 6.244 L^3$ ,*

$$\lambda_{\min}(Q_\delta) \geq \lambda^*(L) - c(L)\delta^2 - \frac{16}{3}L^5\delta^4 e^{2L\delta}.$$

*This is unconditional exactly where Weil positivity on the window has been certified: for  $L \leq 0.8$ , since Zhu [11] certifies  $\lambda^*(0.8) \geq 8.9 \times 10^{-18}$  and  $\lambda^*$  is non-increasing in  $L$ . For  $L > 0.8$  it holds with  $\lambda^*(L)$  replaced by 0, using no positivity input at all.*

**Theorem 8** (sharper, conditional). [T1] *Suppose every nontrivial zero other than the one being moved lies on the critical line, so that  $\tilde{Q} := Q_0 - 4F(\gamma)^2 \geq 0$ . Completing the square,  $4F^2 - 4\delta^2 FF'' = (2F - \delta^2 F'')^2 - \delta^4 F''^2$ , removes the  $FF''$  allowance into  $O(\delta^4)$  and gives*

$$\lambda_{\min}(Q_\delta) \geq -4\delta^2 \sup_{\gamma} \frac{|F'(\gamma)|^2}{\|f\|^2} - \left(\frac{2}{5} + \frac{16}{3}e^{2L\delta}\right)L^5\delta^4 \geq -\frac{8L^3}{3}\delta^2 - O(\delta^4).$$

*Remark 9* (regime — part of the statement). As  $\delta \rightarrow 0$ ,  $\lambda_{\min}(Q_\delta) \rightarrow \lambda^*(L) > 0$ : the form returns to the honest positive floor. The law  $\lambda_{\min} \approx -C(L)\delta^2$  describes only the regime  $C(L)\delta^2 \gg \lambda^*(L)$ . Moreover  $C(L)$  is a property of the *re-optimised* minimiser; evaluating (3) at the unperturbed ground state is 3 to 6 orders of magnitude too small, because  $Q_0$  has a near-null space with  $\lambda_2/\lambda_1 \sim 10^6$ .

### 4.3 The constant is nearly attained

[T2, src/task34\_extra.py, results/stage3\_C\_vs\_CS.csv]

| $L$             | 0.8    | 1.0    | 1.3    | 1.6    | 2.0          |
|-----------------|--------|--------|--------|--------|--------------|
| measured $C(L)$ | 0.3108 | 0.8530 | 2.4987 | 4.9218 | 10.083       |
| $C/(8L^3/3)$    | 0.228  | 0.320  | 0.427  | 0.451  | 0.473        |
| $C/(4L^3/3)$    | 0.455  | 0.640  | 0.853  | 0.901  | <b>0.945</b> |

Keeping the  $\sin^2$  weight in Cauchy–Schwarz sharpens  $|F'(\gamma)|^2 \leq \int u^2 \sin^2(\gamma u) du \rightarrow L^3/3$  as  $\gamma \rightarrow \infty$ , giving the constant  $4L^3/3$ ; the re-optimised minimiser reaches 94.5% of it at  $L = 2$  and is still rising.

### 4.4 One law, two settings

[T3] Identifying the window *length* ( $R+1$  lattice points  $\leftrightarrow$  an interval of length  $2L$ , i.e.  $R \leftrightarrow 2L$ ) and the pair-versus-quartet factor 2,

$$\text{function field: } C_R = \frac{R^3}{6} + O(R^2), \quad \text{zeta: } C(L) = \frac{4L^3}{3} = \frac{(2L)^3}{6},$$

the same expression (window length)<sup>3</sup>/6.

## 5 The $\zeta$ side as a control

[T2; src/stage3\_converge.py, src/stage3\_connes.py]

We assemble the window form from Zhu’s geometric side in three closed-form bases (two even, one odd), moving everything to  $u$ -space by Parseval so no oscillatory quadrature is needed, and summing the archimedean  $m$ -series analytically via digamma, Lerch  $\Phi$  and Hurwitz  $\zeta$ . The odd-sector pole term  $-2P_j P_k$  — the sign flips because  $F(-i/2) = -F(i/2)$  for odd  $f$  — is derived here.

**Gate A.** Under RH the explicit formula gives  $Q(f) = \sum_\rho |F(\gamma_\rho)|^2$ . The assembled geometric side matches the zeros side to  $1.63 \times 10^{-7}$  (even, vanishing at  $\pm L$ ),  $2.26 \times 10^{-7}$  (odd) and  $4.23 \times 10^{-3}$  (full cosine, which jumps at  $\pm L$ ) at  $K = 800$  zeros, each decaying at the rate that basis’s smoothness predicts ( $\gamma_K^{-3}$  versus  $\gamma_K^{-1}$ ). *Control*: flipping the pole sign inflates the discrepancy to 5.35 and 0.212, so the test discriminates.

**Gate B.**  $\lambda_{\min}(N)$  decreases monotonically in two independent complete even bases to  $2.2702 \times 10^{-17}$  and  $2.3357 \times 10^{-17}$  at  $N = 24$ , landing on Zhu’s certified  $[8.9 \times 10^{-18}, 2.27 \times 10^{-17}]$ . Zhu normalises by  $Q(f)/\|f\|^2$ , the same normalisation used here. We report the inequality, not the matching digits: the sequence is still falling.

**Connes’s window.** At  $2L = \log 13$ ,  $N = 36$ , 140 digits:  $\lambda_{\min} = 8.977 \times 10^{-52}$ , simple and even, with the ground state’s Fourier zeros matching  $\gamma_1$  to  $7.2 \times 10^{-48}$  and tracking roughly  $N/2$  leading zeros. This reproduces at smaller scale what Groskin [6] computes at  $N = 100$ –250.

**Simplicity and evenness.** [T2,  $N = 12, 40$  digits] At  $L \in \{0.5, 0.8, 1.0, \frac{1}{2} \log 13\}$  the ground state is simple with relative gap  $2 \times 10^4$  to  $4 \times 10^5$  and *even* in every case, the odd sector sitting 2–5 orders above. The  $L = 0.8$  entry is consistent with Zhu’s certificate; the other three carry no external certificate and are T2 only.

## 6 Detection onset

[T2; src/stage3e\_certify.py, results/stage3e\_certified\_onset.csv]

**Precedent.** Bombieri [1, §13] already ran this experiment in 2000: the first  $N \leq 160$  zeros of  $\zeta$  together with “a fictitious zero  $\rho_0$  off the critical line, together with their images by complex conjugation and reflection” — the same quartet — at  $\rho_0 = 0.52 + 3.14i$ , on windows  $E = [-t, t]$ , and found a critical value  $t_c$  beyond which the negative eigenvalue stays bounded away from 0. He also separates even and odd eigenfunctions. Our contribution is to resolve that critical window in  $\delta$  and in basis size, and to identify it with Zhu’s bandwidth  $T^*(L) = 2\pi e^{2L}$ .

**The certification asymmetry.** A negative Rayleigh quotient from *any* trial function certifies indefiniteness. A finite basis that finds none certifies nothing. We therefore grow the basis (low block  $\cup$  resonant block  $\{k : w_k \approx \gamma\}$ ) and report, for each planted ordinate, the smallest window at which *some* trial function goes negative. The geometric side supplies the on-line part exactly; truncating the zeros sum omits positive mass  $\sim 10^{-6}$  against a signal  $\sim 10^{-27}$  and cannot certify anything near the onset.

| planted at    | $\gamma_n$ | $L_{\text{pred}} = \frac{1}{2} \log \frac{\gamma}{2\pi}$ | onset, small basis | onset, largest basis |
|---------------|------------|----------------------------------------------------------|--------------------|----------------------|
| $\gamma_1$    | 14.13      | 0.405                                                    | 1.3×               | 1.3×                 |
| $\gamma_5$    | 32.94      | 0.828                                                    | 1.0×               | 1.0×                 |
| $\gamma_{10}$ | 49.77      | 1.035                                                    | not found          | 1.0×                 |
| $\gamma_{30}$ | 101.3      | 1.390                                                    | not found          | <b>0.85×</b>         |
| $\gamma_{80}$ | 201.3      | 1.733                                                    | not found          | 1.15×                |

The onset tracks  $L_{\text{pred}}$  within about  $\pm 30\%$  and moves *earlier* as the basis grows. Detection needs both  $L \gtrsim L_{\text{pred}}(\gamma_n)$  (the bandwidth must reach the ordinate) and  $N \gtrsim 2n$  (the basis must be able to annihilate the first  $n$  zeros) — the second being the same  $N/2$  rule seen in §5. Rows whose magnitude falls below the measured noise floor (the geometric control is PSD by construction, so a negative value there *is* the floor) are excluded and flagged in the data file.

**Weak  $\delta$ -dependence, explained.** Detection is  $C(L)\delta^2 > \lambda^*(L)$ . By Zhu’s Landau–Widom law  $\lambda^*(L)$  falls super-exponentially in  $L$  while  $C(L) \sim L^3$  grows polynomially, so a factor in  $\delta^2$  shifts the crossing only logarithmically. Measured: a tenfold change in  $\delta$  moves the onset from  $L = 0.5$  to  $L = 0.6$ .

## 7 The recorder split

[T2; src/task2\_inertia\_reconcile.py]

Write  $T_R = A - 2M$ . In the function field the pole part of  $A$  is  $q^{|n|/2} + q^{-|n|/2} = u_i w_j + w_i u_j$ , an exact rank-2 *hyperbolic* plane whose scale grows like  $q^{R/2}$  against a fixed  $2g \cdot I$ ; so  $A$  is eventually indefinite under both conventions tested,  $M$  is indefinite, and  $S = T_R$  is PSD. This is structural.

On  $\zeta$  windows it is not. With  $A = \text{Pole} + \text{Arch} - \log \pi \cdot I$ ,  $2M = \text{Prime}$  one finds  $A$  with negative index 1 for  $L \leq 0.8$  and 2 at  $L = \frac{1}{2} \log 13$ , identical in all three bases. But grouping

$-\log \pi$  with the primes instead makes **both**  $A$  and  $M$  **positive definite** at  $L \leq 0.8$ . The reason is structural: for even  $f$  the  $\zeta$  pole term is  $h(i/2) + h(-i/2) = 2F(i/2)^2 \geq 0$ , rank one and *definite*, not hyperbolic. There is no hyperbolic plane on the  $\zeta$  side to find. We therefore state the convention ( $A = \text{Pole} + \text{Arch} - \log \pi \cdot I$ ) and claim Block F only for the function field.

## 8 Relation to prior work

**Hallouin–Perret** [8] is the function-field foundation: Proposition 5 is the window Toeplitz matrix, Theorem 6 the first-singular-size and kernel generator, Theorem 36(ii) step (i), Lemmas 33–34 the rank rule and parity split used here. **Connes–van Suijlekom** [5] supply the  $C^*$ -algebraic proof of step (i) and the Hurwitz step, cite [8] for the function-field connection, and state the simplicity difficulty themselves; their Remark 2.3 is our degeneracy observation in words. **Bombieri** [1] is the precedent for planted  $\zeta$  experiments (§13) and proves the negative-eigenvalue count (Theorem 8); **Yoshida** [10] first studied the Weil hermitian form on functions supported in  $[-t, t]$ . **Zhu** [11] supplies the certified  $\lambda^*(0.8)$ , the normalisation and the bandwidth  $T^*$ . **Groskin** [6, 7] supplies the large-scale Galerkin computations and the archimedean tail order. **Connes–Consani–Moscovici** [3], **Connes** [4], **Connes–Consani** [2] and **Suzuki** [9] are the surrounding program. A fuller account, with verbatim quotations and the verification method, is in `PRIOR_ART.md`.

## 9 Limitations

1. No rigorous lower bound on  $\lambda^*(L)$  of our own; Theorem 7 is unconditional only where Zhu’s certificate reaches ( $L \leq 0.8$ ).
2. Theorem 8 is conditional on all other zeros being on the line.
3. Interval certification covers the zeros-side assembly only: `mpmath`’s interval context has no digamma and no Lerch transcendent, so the geometric side could not be interval-evaluated. What is certified instead is the  $\text{LDL}^\top$  witness identity  $v^\top Z v = D_j$  and a measured noise floor.
4. Onset results are limited by  $N \gtrsim 2n$ ; the largest subspace used had dimension 59.
5.  $C(L)$  is measured at finite  $\delta$  and finite basis; the saturation  $C/(4L^3/3) \rightarrow 1$  is observed, not proved.
6. The  $\zeta$ -side numbers depend on the planted ordinate, the basis and the truncation.

## 10 Traps this project documents

The retraction log (Appendix A) is part of the result. The reusable lessons: **(a)** truncating the zeros sum biases the form *downward* and manufactures false detections; **(b)** moving a zero and adding a zero are two different counterfactuals, both legitimate — Bombieri studies the second — and the sharp law needs the first; **(c)** `float64` returns spurious negative eigenvalues of size  $10^{-16}$  where the true value is exactly 0; **(d)** interval libraries have gaps, and `mpmath`’s `iv.sinapi` is itself broken (it calls a non-existent `ctx.sinpi`); **(e)** a non-detection certifies nothing; **(f)** keeping known-good regression values was the only thing that caught three sector-rename bugs.

## A Retractions

Sixteen corrections were logged during this work; the full list with evidence is in `REPORT.md`. Those a reader must see:

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| #   | Retraction                                                                                                                                                                                                                                                                                                                                                             |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Frobenius radius computed on $P$ rather than its reverse ( $0.8$ instead of $\leq 1.3 \times 10^{-51}$ ).                                                                                                                                                                                                                                                              |
| 3   | A tolerance-based sign test misclassified exact zeros; replaced by an error-based test plus exact rational certificates.                                                                                                                                                                                                                                               |
| 4   | <b>H2 as first stated was refuted by my own adversarial pass</b> ; restated with a distinctness hypothesis. The restatement is Theorem 2.                                                                                                                                                                                                                              |
| 6   | <b>“Step (iv) is free” retracted</b> — it had been scored on zero angles, which is the wrong object. At the function level it carries content (Proposition 5).                                                                                                                                                                                                         |
| 8–9 | A 40-digit run discarded; three sector-rename bugs caught only by regression values.                                                                                                                                                                                                                                                                                   |
| 11  | <b>“Low-lying zeros only” retracted as structural</b> . The height sweep had used a truncated on-line side, and non-detections were basis-limited. Redone, $\gamma_{30} \approx 101$ is detected at $0.85 L_{\text{pred}}$ . <i>Amended</i> : the added-quartet counterfactual is legitimate in its own right — it is Bombieri’s — and was not the error.              |
| 12  | $\gamma_{80}$ “detections” below their own noise floor discarded; later detected at $1.15 L_{\text{pred}}$ with a positive control.                                                                                                                                                                                                                                    |
| 15  | <b>“Block F holds on the <math>\zeta</math> side” retracted</b> (§7).                                                                                                                                                                                                                                                                                                  |
| 16  | <b>Three claimed contributions are published theorems</b> : step (i) is [8, Thm. 36(ii)], the first-singular-size result is [8, Thm. 6], the window matrix is [8, Prop. 5]; the degeneracy observation is [5, Rem. 2.3]; the negative-index count is [1, Thm. 8]; the planted experiment is [1, §13]. An earlier abstract-level novelty check had missed all of these. |

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## B Reproduction

Code and data: <https://github.com/innerlightr-wq/weil-window-control>. Python 3.12, mpmath 1.3.0, numpy 1.26.4, matplotlib 3.8+; no gmpy2 (installing it speeds the high-precision runs substantially).

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| Result                | Script                         | Data                           |
|-----------------------|--------------------------------|--------------------------------|
| §3 genuine            | src/stage1.py                  | stage1_spectra.csv             |
| §3 planted            | src/stage2.py                  | stage2_spectra.csv             |
| §3 kernel             | src/function_level.py          | stage2_function_level.csv      |
| Thm 6, 7, 8           | src/proofs_check.py            | stage3_sharp_bound.csv         |
| $C$ vs Cauchy–Schwarz | src/task34_extra.py            | stage3_C_vs_CS.csv             |
| Gates A, B            | src/stage3_converge.py         | stage3_gateA/B_*.csv           |
| Connes window         | src/stage3_connes.py           | stage3b_connes_zeros_N36.csv   |
| Detection onset       | src/stage3e_certify.py         | stage3e_certified_onset.csv    |
| Recorder split        | src/task2_inertia_reconcile.py | stage3d_inertia_reconciled.csv |

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